Electrical Trainer Kit



Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium**

3 labs

Covers these lab items: Kirchhoff's Law kit

What it is used for

To demonstrate and verify Kirchhoff's current and voltage laws using an electrical trainer kit. To make safe circuit connections and compare measured values with calculated results.

Main parts

- Trainer board or circuit panel
- DC power supply
- Resistors or load components
- Connecting leads
- Ammeter or multimeter
- Voltmeter or multimeter
- Switches and terminals

Before you start

1. Read the trainer manual and identify the circuit diagram.
2. Switch off the supply before making or changing connections.
3. Inspect leads, terminals, meters, and components for damage.
4. Confirm the supply voltage and component ratings.
5. Select the correct meter function, range, and polarity.

How to use

1. Build the specified circuit with the power supply switched off.
2. Check all connections against the circuit diagram.
3. Set the meters to the required current or voltage measurement mode.
4. Switch on the supply and increase it only to the specified level.
5. Measure branch currents and voltage drops at the designated points.
6. Record readings and compare current sums and voltage sums with Kirchhoff's laws.
7. Switch off the supply before changing the circuit or removing leads.

⚠ Safety

- Never connect an ammeter directly across a voltage source.
- Do not exceed component or meter ratings.
- Avoid touching exposed conductive parts when energized.
- Report overheating, smoke, or abnormal readings and switch off the supply.

Care & storage

- Switch off and disconnect the supply after use.
- Return leads and components to their labelled locations.
- Inspect meters and leads for damage after the exercise.
- Keep the trainer board clean and dry.

Common mistakes to avoid

- Changing wiring while the supply is on.
- Using the wrong meter terminal or range.
- Reversing polarity where it matters.
- Recording readings without checking connections and instrument settings.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.